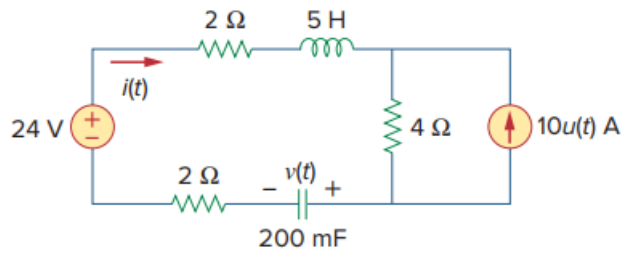


Problem

Obtain $v(t)$ and $i(t)$ for $t > 0$ in the circuit of Fig. 8.84.

Figure 8.84:



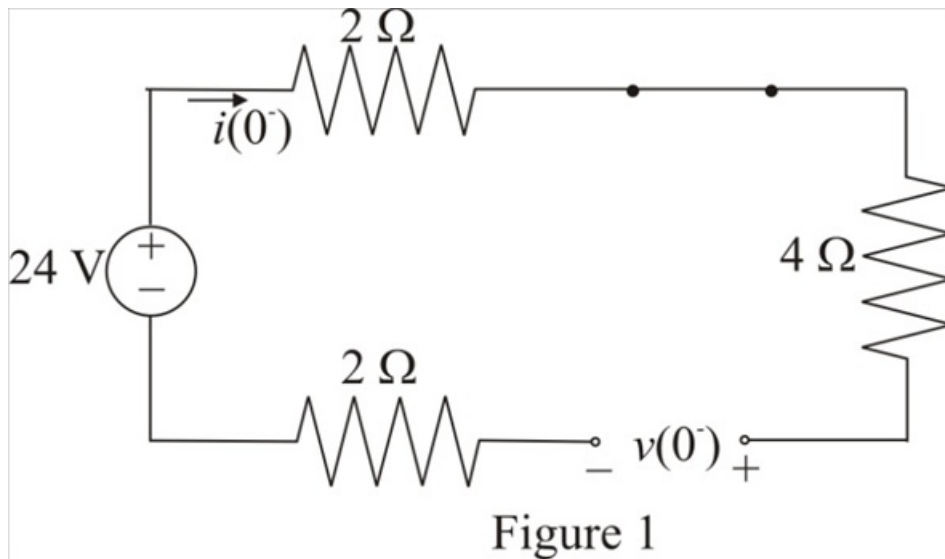
Step-by-step solution

Step 1 of 11

Refer to Figure 8.84 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit. The value of the current source is zero since $u(t) = 0$ for $t < 0$. That is the current source acts as open circuit $t < 0$.

Draw the circuit diagram at $t < 0$.



Calculate the voltage across capacitor for $t < 0$.

$$\begin{aligned} v(0^-) &= v_s(0^-) \\ &= 24 \text{ V} \end{aligned}$$

From Figure 1, as the circuit is an open circuit.

$$i(0^-) = 0 \text{ A}$$

The values of the current $i(0^-)$ and the voltage $v(0^-)$ are 0 A and 24 V respectively.

Step 2 of 11

Since $u(t)=1$ for $t \geq 0$, the value of the current source is 10 A .

Draw the circuit diagram at $t > 0$.

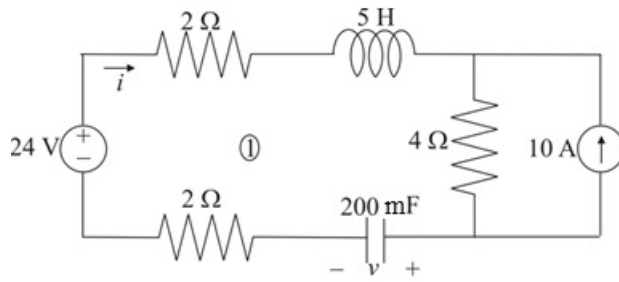


Figure 2

[Comments \(1\)](#)



Anonymous

How do you know $u(t)=1$ for $t \geq 0$?

Step 3 of 11

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 24 V and the value of the current $i(0^+)$ is 0 A .

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 24 \text{ V}$ and $i(0) = 0 \text{ A}$.

Step 4 of 11

Transform the current source to voltage source with a 4Ω resistor in series.

The value of the voltage source will be $4 \times 10 = 40 \text{ V}$.

Sketch the modified circuit as shown in Figure 3.

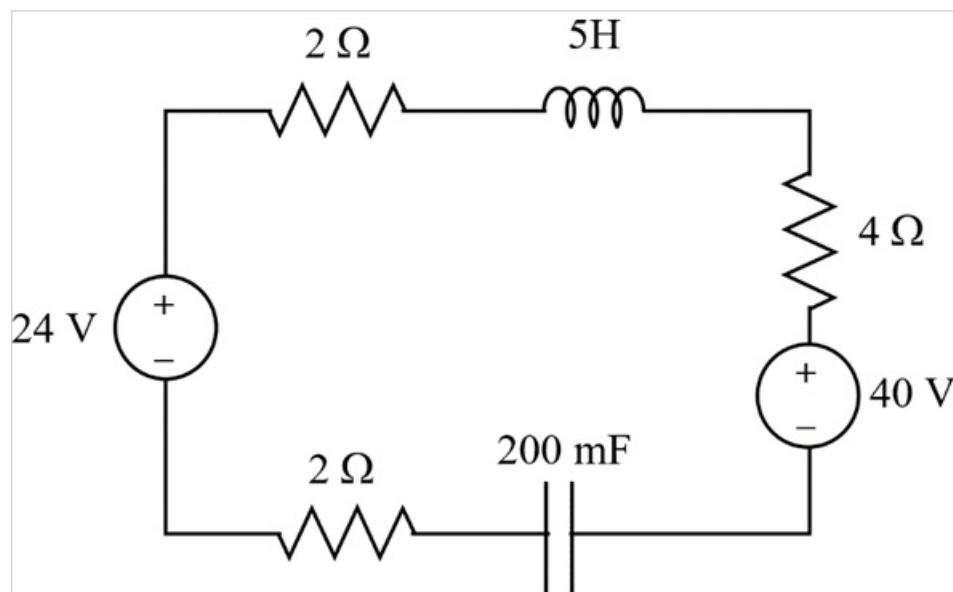


Figure 3

Step 5 of 11

Observe that the three resistors are in series and the circuit is a series RLC circuit.

Calculate the equivalent supply voltage.

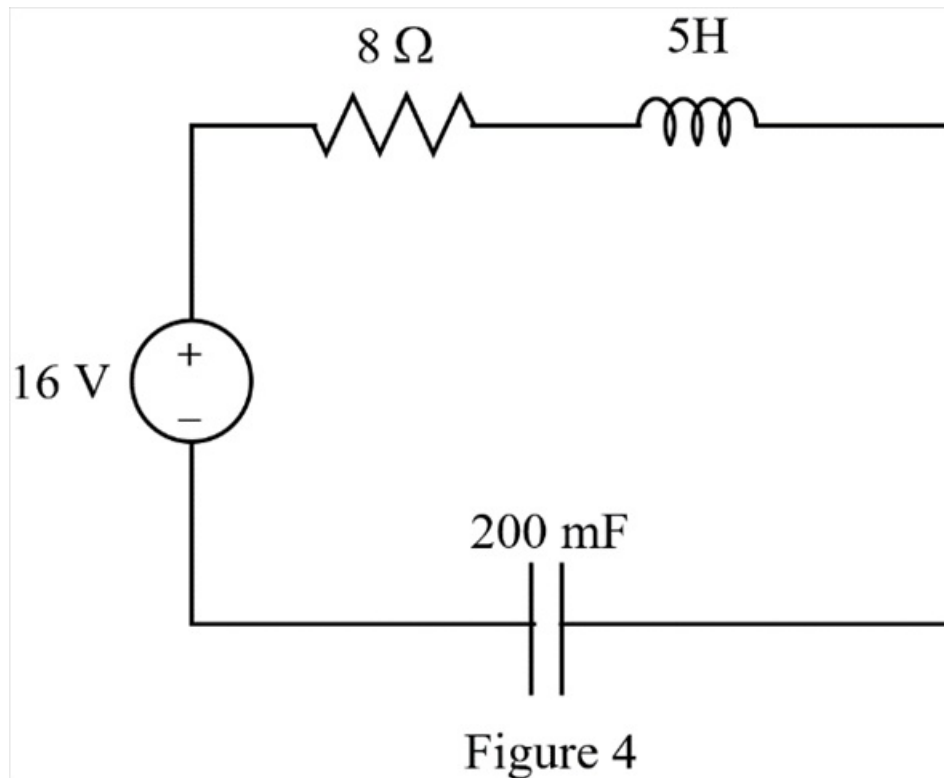
$$\begin{aligned}V_s &= 40 - 24 \\&= 16 \text{ V}\end{aligned}$$

Calculate the equivalent resistance.

$$\begin{aligned}R &= 2 + 2 + 4 \\&= 8 \Omega\end{aligned}$$

Step 6 of 11

Sketch the modified circuit as shown in Figure 4.



Step 7 of 11

Figure 4 shows a series *RLC* circuit.

Determine the characteristic damping ratio, α .

$$\begin{aligned}\alpha &= \frac{R}{2L} \\&= \frac{8}{(2)(5)} \\&= 0.8\end{aligned}$$

Determine the natural frequency of oscillation, ω_o .

$$\begin{aligned}\omega_o &= \frac{1}{\sqrt{LC}} \\&= \frac{1}{\sqrt{(5)(200 \times 10^{-3})}} \\&= 1 \text{ rad/s}\end{aligned}$$

Step 8 of 11

Since $\alpha < \omega_0$, the case is underdamped, the voltage across the capacitor is expressed as,

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \dots\dots (1)$$

Calculate the damping frequency, ω_d .

$$\begin{aligned}\omega_d &= \omega_n \sqrt{1 - \alpha^2} \\ &= (1) \left(\sqrt{1 - 0.8^2} \right) \\ &= 0.6\end{aligned}$$

Step 9 of 11

Substitute 0.8 for α , 0.6 rad/s for ω_d , 16 V for V_s in (1).

$$v(t) = 16 + (A_1 \cos 0.6t + A_2 \sin 0.6t) e^{-0.8t}$$

At $t = 0$, $v(0) = 24$ V.

$$24 = 16 + (A_1 + A_2(0))(1)$$

$$A_1 = 8$$

Thus,

$$v(t) = 16 + (8 \cos 0.6t + A_2 \sin 0.6t) e^{-0.8t} \dots\dots (2)$$

Step 10 of 11

Differentiate (2) with respect to t .

$$\begin{aligned}\frac{dv(t)}{dt} &= 0 + ((-8 \sin 0.6t)(0.6) + (A_2 \cos 0.6t)(0.6)) e^{-0.8t} + (-0.8)(8 \cos 0.6t + A_2 \sin 0.6t) e^{-0.8t} \\ &= (0.6A_2 \cos 0.6t - 4.8 \sin 0.6t - 6.4 \cos 0.6t - 0.8A_2 \sin 0.6t) e^{-0.8t} \\ &= ((0.6A_2 - 6.4) \cos 0.6t - (4.8 + 0.8A_2) \sin 0.6t) e^{-0.8t}\end{aligned}$$

Calculate the current through the capacitor, $i(t)$.

$$\begin{aligned}i(t) &= C \frac{dv(t)}{dt} \\ i(t) &= (200 \times 10^{-3}) (((0.6A_2 - 6.4) \cos 0.6t - (4.8 + 0.8A_2) \sin 0.6t) e^{-0.8t}) \\ i(t) &= 0.2 (((0.6A_2 - 6.4) \cos 0.6t - (4.8 + 0.8A_2) \sin 0.6t) e^{-0.8t}) \dots\dots (3)\end{aligned}$$

At $t = 0$, $i(0) = 0$.

$$0 = 0.2 (((0.6A_2 - 6.4)(1) - (4.8 + 0.8A_2)(0))(1))$$

$$0 = 0.2(0.6A_2 - 6.4)$$

$$A_2 = 10.667$$

Step 11 of 11

Rewrite equation (2) as:

$$v(t) = 16 + (8 \cos 0.6t + 10.667 \sin 0.6t) e^{-0.8t} \text{ V}$$

Rewrite equation (3) as:

$$\begin{aligned} i(t) &= 0.2 \left(\left((0.6(10.667) - 6.4) \cos 0.6t - (4.8 + 0.8(10.667)) \sin 0.6t \right) e^{-0.8t} \right) \\ &= 0.2 (-13.33 \sin 0.6t) e^{-0.8t} \\ &= -2.667 \sin 0.6t e^{-0.8t} \text{ A} \end{aligned}$$

Therefore, the voltage $v(t)$ is $16 + (8 \cos 0.6t + 10.667 \sin 0.6t) e^{-0.8t} \text{ V}$ and the current $i(t)$ is $-2.667 \sin 0.6t e^{-0.8t} \text{ A}$.